NOVOSENSE
纳芯微电子

NSi826xC

High Reliability Reinforced Six-Channel Digital Isolators

Datasheet (EN) 1.2

Product Overview

The NSi826xC devices are high reliability six-channel digital isolators. The NSi826xC device is safety certified by UL1577 support several insulation withstand voltage(5kV), while providing high electromagnetic immunity and low emissions at low power consumption. The data rate of the NSi826xC is up to 150Mbps, and the common-mode transient immunity (CMTI) is up to 250kV/us. The NSi826xC device provides digital channel direction configuration and the default output level configuration when the input power is lost. Wide supply voltage of the NSi826xC device supports to connect with most digital interface directly, easy to do the level shift. High system level EMC performance enhance reliability and stability of use.

Key Features

- Up to 5000V_{rms} Insulation voltage
- Data rate: DC to 150Mbps
- Power supply voltage: 2.5V to 5.5V
- High CMTI: 250kV/us
- Chip level ESD: HBM: ±8kV
- High system level EMC performance:
  - Enhanced system level ESD, EFT, Surge immunity
- Default output high level or low level option
- Low power consumption: 1.5mA/ch (1 Mbps)
- Low propagation delay: <15ns
- Operation temperature: -40℃~125℃
- RoHS-compliant packages:
  - SOP16(300mil)

Safety Regulatory Approvals

- UL recognition: up to 5000V_{rms} for 1 minute per UL1577
- CQC certification per GB4943.1-2022
- CSA component notice 5A
- DIN EN IEC 60747-17 (VDE 0884-17)

Applications

- Industrial automation system
- Isolated SPI, RS232, RS485
- General-purpose multichannel isolation

Device Information

Part Number	*Package*	*Body Size*
NSi826xC-DSWR	SOP16(300mil)	10.30mm × 7.50mm

Functional Block Diagrams

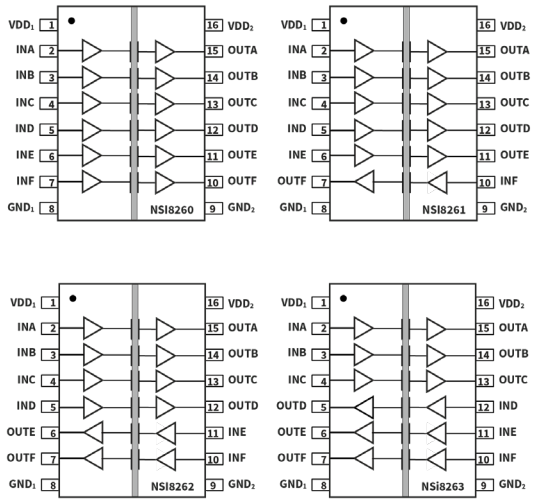


Figure 1. NSi826xC Block Diagram

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1. Pin Configuration and Functions

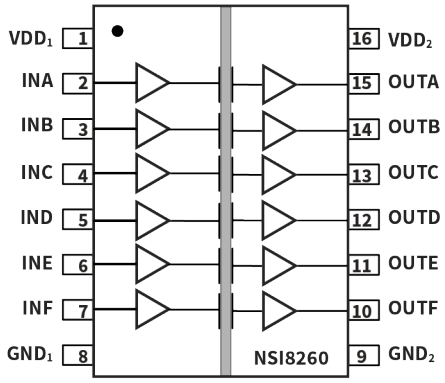


Figure 1.1 NSi8260 SOP16 Package

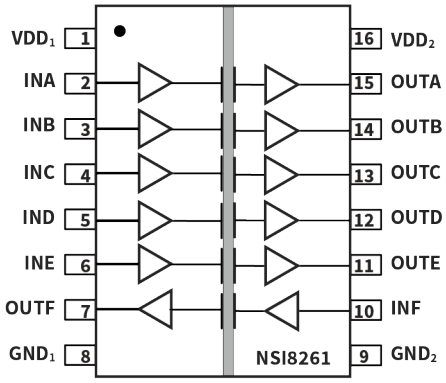


Figure 1.2 NSi8261 SOP16 Package

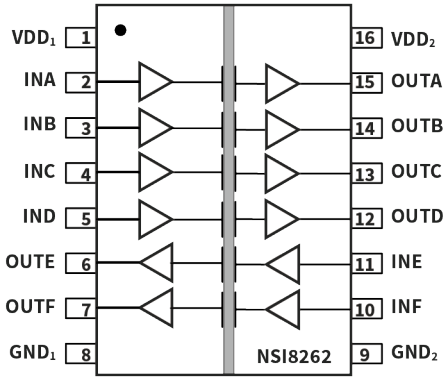


Figure 1.3 NSi8262 SOP16 Package

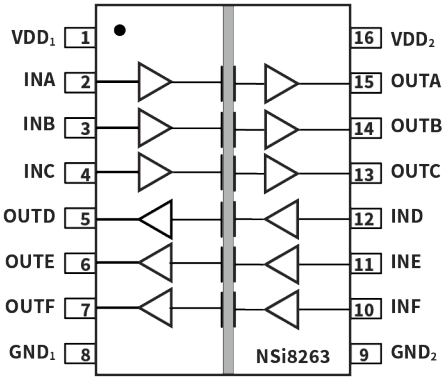


Figure 1.4 NSi8263 SOP16 Package

Table 1.1 NSi8260/ NSi8261/ NSi8262 SOP16 Pin Configuration and Description

NSi8260 PIN NO.	*NSi8261 PIN NO.*	*NSi8262 PIN NO.*	*NSi8263 PIN NO.*	*SYMBOL*	*FUNCTION*
1	1	1	1	VDD_1	Power Supply for Isolator Side 1
2	2	2	2	INA	Logic Input A
3	3	3	3	INB	Logic Input B
4	4	4	4	INC	Logic Input C
5	5	5	12	IND	Logic Input D
6	6	11	11	INE	Logic Input E
7	10	10	10	INF	Logic Input F
8	8	8	8	GND_1	Ground 1, the ground reference for Isolator Side 1

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NSi8260 PIN NO.	NSi8261 PIN NO.	NSi8262 PIN NO.	NSi8263 PIN NO.	SYMBOL	FUNCTION
9	9	9	9	GND_2	Ground 2, the ground reference for Isolator Side 2
10	7	7	7	OUTF	Logic Output F
11	11	6	6	OUTE	Logic Output E
12	12	12	5	OUTD	Logic Output D
13	13	13	13	OUTC	Logic Output C
14	14	14	14	OUTB	Logic Output B
15	15	15	15	OUTA	Logic Output A
16	16	16	16	VDD2	Power Supply for Isolator Side 2

2. Absolute Maximum Ratings

Parameters	Symbol	Min	Typ	Max	Unit	Comments
Power Supply Voltage	VDD1, VDD2	-0.5		6.5	V	
Maximum Input Voltage	VINA, VINB, VINC, VIND, VINE, VINF	-0.4		VDD+0.4	V	
Maximum Output Voltage	VOUTA, VOUTB, VOUTC, VOUTD, VOUTE, VOUTF	-0.4		VDD+0.4	V	
Maximum Input/Output Pulse Voltage	ALL I/O Pin	-0.8		VDD+0.8	V	Pulse width should be less than 100ns, and the duty cycle should be less than 10%
Output current	I_o	-15		15	mA	
Maximum Surge Isolation Voltage	V_{IOSM}			6.25	kV	
Operating Temperature	T_{opr}	-40		125	℃	
Junction Temperature	T_j			150	℃	
Storage Temperature	T_{stg}	-65		150	℃	
Electrostatic discharge	HBM			±8000	V	
	CDM			±2000	V	

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3. Recommended Operating Conditions

Parameters	*Symbol*	*min*	*typ*	*max*	*unit*
Power Supply Voltage	VDD1, VDD2	2.5		5.5	V
Operating Temperature	T_{opr}	-40		125	℃
High Level Input Voltage	VIH	2			V
Low Level Input Voltage	VIL			0.8	V
Data rate	DR			150	Mbps

4. Thermal Characteristics

Parameters	*Symbol*	*SOP16(300mil)*	*Unit*
IC Junction-to-Air Thermal Resistance	θ_{JA}	60.3	°C/W
Junction-to-case (top) thermal resistance	$\theta_{JC(top)}$	24.0	°C/W
Junction-to-board thermal resistance	θ_{JB}	29.3	°C/W

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5. Specifications

5.1. Electrical Characteristics

(VDD1=2.5V~5.5V, VDD2=2.5V~5.5V, T_A=-40℃ to 125℃. Unless otherwise noted, Typical values are at VDD1 = 5V, VDD2 = 5V, T_A = 25℃)

Parameters	*Symbol*	*Min*	*Typ*	*Max*	*Unit*	*Comments*
Power on Reset	VDD_{POR}		2.2		V	POR threshold as during power-up
	VDD_{HYS}		0.1		V	POR threshold Hysteresis
Input Threshold	V_{IT}		1.6		V	Input Threshold at rising edge
	V_{IT_HYS}		0.4		V	Input Threshold Hysteresis
High Level Input Voltage	V_{IH}	2			V	
Low Level Input Voltage	V_{IL}			0.8	V	
High Level Output Voltage	V_{OH}	VDD-0.4			V	I_{OH} =- 4mA
Low Level Output Voltage	V_{OL}			0.4	V	I_{OL} = 4mA
Output Impedance	R_{out}		50		ohm	
Input Pull high or low Current	I_{pull}		8	15	μA	
Start Up Time after POR	t_{rbs}		10		μs	
Common Mode Transient Immunity	CMTI	±200	±250		kV/μs	See Figure 5.12 , C_L = 15pF

5.2. Supply Current Characteristics – 5V Supply

(VDD1=5V± 10%, VDD2=5V± 10%, T_A=-40℃ to 125℃. Unless otherwise noted, Typical values are at **VDD1 = 5V, VDD2 = 5V,** T_A = 25℃)

Parameters	*Symbol*	*Min*	*Typ*	*Max*	*Unit*	*Comments*
Supply current	**NSi8260**					
	I_{DD1}(Q0)		1.39	3.09	mA	All Input 0V for NSi8260x0 Or All Input at supply for NSi8260x1
	I_{DD2}(Q0)		3.41	5.63	mA	
	I_{DD1}(Q1)		7.37	12.16	mA	All Input at supply for NSi8260x0 Or All Input 0V for NSi8260x1
	I_{DD2}(Q1)		3.49	5.76	mA	

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Parameters	Symbol	Min	Typ	Max	Unit	Comments
	I_{DD1}(1M)		4.39	7.24	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.67	6.06	mA	
	I_{DD1}(10M)		4.71	7.77	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		5.66	9.34	mA	
	I_{DD1}(100M)		7.47	14.94	mA	All Input with 100Mbps, C_L=15pF
	I_{DD2}(100M)		23.8	55.22	mA	
	NSi8261					
	I_{DD1}(Q0)		1.73	2.85	mA	All Input 0V for NSi8261x0 Or All Input at supply for NSi8261x1
	I_{DD2}(Q0)		3.07	5.07	mA	
	I_{DD1}(Q1)		6.72	11.09	mA	All Input at supply for NSi8261x0 Or All Input 0V for NSi8261x1
	I_{DD2}(Q1)		4.14	6.83	mA	
	I_{DD1}(1M)		4.27	7.05	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.79	6.25	mA	
	I_{DD1}(10M)		4.87	8.03	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		5.50	9.08	mA	
	I_{DD1}(100M)		10.19	20.38	mA	All Input with 100Mbps, C_L= 15pF
	I_{DD2}(100M)		21.08	47.21	mA	
	NSi8262					
	I_{DD1}(Q0)		2.06	3.40	mA	All Input 0V for NSi8262x0 Or All Input at supply for NSi8262x1
	I_{DD2}(Q0)		2.74	4.52	mA	
	I_{DD1}(Q1)		6.08	10.03	mA	All Input at supply for NSi8262x0 Or All Input 0V for NSi8262x1
	I_{DD2}(Q1)		4.78	7.89	mA	
	I_{DD1}(1M)		4.15	6.85	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.91	6.45	mA	
	I_{DD1}(10M)		5.03	8.29	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		5.34	8.82	mA	
	I_{DD1}(100M)		12.91	25.83	mA	All Input with 100Mbps, C_L= 15pF
	I_{DD2}(100M)		18.36	39.2	mA	

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Parameters	Symbol	Min	Typ	Max	Unit	Comments
	NSi8263					
	I_{DD1}(Q0)		2.40	3.96	mA	All Input 0V for NSi8263x0 Or All Input at supply for NSi8263x1
	I_{DD2}(Q0)		2.40	3.96	mA	
	I_{DD1}(Q1)		5.43	8.96	mA	All Input at supply for NSi8263x0 Or All Input 0V for NSi8263x1
	I_{DD2}(Q1)		5.43	8.96	mA	
	I_{DD1}(1M)		4.03	6.65	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		4.03	6.65	mA	
	I_{DD1}(10M)		5.19	8.56	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		5.19	8.56	mA	
	I_{DD1}(100M)		15.64	31.27	mA	All Input with 100Mbps, C_L= 15pF
	I_{DD2}(100M)		15.64	31.27	mA	

5.3. Supply Current Characteristics –3.3V Supply

(VDD1=3.3V± 10%, VDD2=3.3V± 10%, T_A=-40℃ to 125℃. Unless otherwise noted, Typical values are at **VDD1 = 3.3V, VDD2 = 3.3V,** T_A = 25℃)

Parameters	Symbol	Min	Typ	Max	Unit	Comments
Supply current	**NSi8260**					
	I_{DD1}(Q0)		1.33	3.00	mA	All Input 0V for NSi8260x0 Or All Input at supply for NSi8260x1
	I_{DD2}(Q0)		3.36	5.54	mA	
	I_{DD1}(Q1)		7.26	11.98	mA	All Input at supply for NSi8260x0 Or All Input 0V for NSi8260x1
	I_{DD2}(Q1)		3.43	5.66	mA	
	I_{DD1}(1M)		4.31	7.11	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.55	5.86	mA	
	I_{DD1}(10M)		4.5	7.43	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		4.87	8.04	mA	
	I_{DD1}(100M)		6.15	12.30	mA	All Input with 100Mbps, C_L=15pF
	I_{DD2}(100M)		18.89	37.78	mA	
	NSi8261					
	I_{DD1}(Q0)		1.67	2.75	mA	

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Parameters	Symbol	Min	Typ	Max	Unit	Comments
	I_{DD2}(Q0)		3.02	4.99	mA	All Input 0V for NSi8261x0 Or All Input at supply for NSi8261x1
	I_{DD1}(Q1)		6.62	10.93	mA	All Input at supply for NSi8261x0 Or All Input 0V for NSi8261x1
	I_{DD2}(Q1)		4.07	6.71	mA	
	I_{DD1}(1M)		4.18	6.90	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.68	6.07	mA	
	I_{DD1}(10M)		4.56	7.53	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		4.81	7.93	mA	
	I_{DD1}(100M)		8.27	16.55	mA	All Input with 100Mbps, C_L= 15pF
	I_{DD2}(100M)		16.77	33.53	mA	
	NSi8262					
	I_{DD1}(Q0)		2.01	3.31	mA	All Input 0V for NSi8261x0 Or All Input at supply for NSi8261x1
	I_{DD2}(Q0)		2.68	4.43	mA	
	I_{DD1}(Q1)		5.98	9.87	mA	All Input at supply for NSi8261x0 Or All Input 0V for NSi8261x1
	I_{DD2}(Q1)		4.71	7.77	mA	
	I_{DD1}(1M)		4.06	6.69	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.80	6.28	mA	
	I_{DD1}(10M)		4.62	7.63	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		4.75	7.83	mA	
	I_{DD1}(100M)		10.40	20.79	mA	All Input with 100Mbps, C_L= 15pF
	I_{DD2}(100M)		14.64	29.29	mA	
	NSi8263					
	I_{DD1}(Q0)		2.35	3.87	mA	All Input 0V for NSi8262x0 Or All Input at supply for NSi8262x1
	I_{DD2}(Q0)		2.35	3.87	mA	
	I_{DD1}(Q1)		5.35	8.82	mA	All Input at supply for NSi8262x0 Or All Input 0V for NSi8262x1
	I_{DD2}(Q1)		5.35	8.82	mA	
	I_{DD1}(1M)		3.93	6.48	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.93	6.48	mA	
	I_{DD1}(10M)		4.69	7.73	mA	All Input with 10Mbps,

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Parameters	Symbol	Min	Typ	Max	Unit	Comments
	I_{DD2}(10M)		4.69	7.73	mA	C_L=15pF
	I_{DD1}(100M)		12.52	25.04	mA	All Input with 100Mbps, C_L = 15pF
	I_{DD2}(100M)		12.52	25.04	mA	

5.4. Supply Current Characteristics–2.5V Supply

(VDD1=2.5V± 10%, VDD2=2.5V± 10%, T_A=-40℃ to 125℃. Unless otherwise noted, Typical values are at VDD1 = 2.5V, VDD2 = 2.5V, T_A = 25℃)

Parameters	Symbol	Min	Typ	Max	Unit	Comments
Supply current	**NSi8260**					
	I_{DD1}(Q0)		1.29	2.94	mA	All Input 0V for NSi8260x0 Or All Input at supply for NSi8260x1
	I_{DD2}(Q0)		3.33	5.49	mA	
	I_{DD1}(Q1)		7	11.55	mA	All Input at supply for NSi8260x0 Or All Input 0V for NSi8260x1
	I_{DD2}(Q1)		3.39	5.59	mA	
	I_{DD1}(1M)		4.17	6.88	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.47	5.73	mA	
	I_{DD1}(10M)		4.29	7.08	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		4.48	7.39	mA	
	I_{DD1}(100M)		5.27	10.54	mA	All Input with 100Mbps, C_L=15pF
	I_{DD2}(100M)		15.33	30.66	mA	
	NSi8261					
	I_{DD1}(Q0)		1.63	2.69	mA	All Input 0V for NSi8261x0 Or All Input at supply for NSi8261x1
	I_{DD2}(Q0)		2.99	4.93	mA	
	I_{DD1}(Q1)		6.40	10.56	mA	All Input at supply for NSi8261x0 Or All Input 0V for NSi8261x1
	I_{DD2}(Q1)		3.99	6.59	mA	
	I_{DD1}(1M)		4.05	6.69	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.59	5.92	mA	
	I_{DD1}(10M)		4.32	7.13	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		4.45	7.34	mA	
	I_{DD1}(100M)		6.95	13.89	mA	All Input with 100Mbps,

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Parameters	Symbol	Min	Typ	Max	Unit	Comments
	I_{DD2}(100M)		13.65	27.31	mA	C_L = 15pF
	NSi8262					
	I_{DD1}(Q0)		1.97	3.25	mA	All Input 0V for NSi8261x0 Or All Input at supply for NSi8261x1
	I_{DD2}(Q0)		2.65	4.37	mA	
	I_{DD1}(Q1)		5.80	9.56	mA	All Input at supply for NSi8261x0 Or All Input 0V for NSi8261x1
	I_{DD2}(Q1)		4.59	7.58	mA	
	I_{DD1}(1M)		3.94	6.50	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.70	6.11	mA	
	I_{DD1}(10M)		4.35	7.18	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		4.42	7.29	mA	
	I_{DD1}(100M)		8.62	17.25	mA	All Input with 100Mbps, C_L = 15pF
	I_{DD2}(100M)		11.98	23.95	mA	
	NSi8263					
	I_{DD1}(Q0)		2.31	3.81	mA	All Input 0V for NSi8262x0 Or All Input at supply for NSi8262x1
	I_{DD2}(Q0)		2.31	3.81	mA	
	I_{DD1}(Q1)		5.20	8.57	mA	All Input at supply for NSi8262x0 Or All Input 0V for NSi8262x1
	I_{DD2}(Q1)		5.20	8.57	mA	
	I_{DD1}(1M)		3.82	6.30	mA	All Input with 1Mbps, C_L=15pF
	I_{DD2}(1M)		3.82	6.30	mA	
	I_{DD1}(10M)		4.39	7.24	mA	All Input with 10Mbps, C_L=15pF
	I_{DD2}(10M)		4.39	7.24	mA	
	I_{DD1}(100M)		10.30	20.60	mA	All Input with 100Mbps, C_L = 15pF
	I_{DD2}(100M)		10.30	20.60	mA	

5.5. Switching Characteristics - 5V Supply

(VDD1=5V± 10%, VDD2=5V± 10%, T_A=-40℃ to 125℃. Unless otherwise noted, Typical values are at **VDD1 = 5V, VDD2 = 5V,** T_A = 25℃)

Parameters	Symbol	Min	Typ	Max	Unit	Comments
Data Rate	DR	0		150	Mbps	

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Parameters	Symbol	Min	Typ	Max	Unit	Comments
Minimum Pulse Width	PW			5.0	ns	
Propagation Delay	t_{PLH}	2.5	6.54	15	ns	See Figure 5.11 , C_L = 15pF
	t_{PHL}	2.5	8.30	15	ns	See Figure 5.11 , C_L = 15pF
Pulse Width Distortion $\lvert t_{PHL} - t_{PLH} \rvert$	PWD			5.0	ns	See Figure 5.11 , C_L = 15pF
Rising Time	t_r			5.0	ns	See Figure 5.11 , C_L = 15pF
Falling Time	t_f			5.0	ns	See Figure 5.11 , C_L = 15pF
Peak Eye Diagram Jitter	t_{JIT}(PK)		350		ps	
Channel-to-Channel Delay Skew	t_{SK}(c2c)			2.5	ns	
Part-to-Part Delay Skew	t_{SK}(p2p)			5.0	ns	

5.6. Switching Characteristics - 3.3V Supply

(VDD1=3.3V± 10%, VDD2=3.3V± 10%, T_A=-40℃ to 125℃. Unless otherwise noted, Typical values are at **VDD1 = 3.3V, VDD2 = 3.3V,** T_A = 25℃)

Parameters	Symbol	Min	Typ	Max	Unit	Comments
Data Rate	DR	0		150	Mbps	
Minimum Pulse Width	PW			5.0	ns	
Propagation Delay	t_{PLH}	2.5	7.5	15	ns	See Figure 5.11 , C_L = 15pF
	t_{PHL}	2.5	8.7	15	ns	See Figure 5.11 , C_L = 15pF
Pulse Width Distortion $\lvert t_{PHL} - t_{PLH} \rvert$	PWD			5.0	ns	See Figure 5.11 , C_L = 15pF
Rising Time	t_r			5.0	ns	See Figure 5.11 , C_L = 15pF
Falling Time	t_f			5.0	ns	See Figure 5.11 , C_L = 15pF
Peak Eye Diagram Jitter	t_{JIT}(PK)		350		ps	
Channel-to-Channel Delay Skew	t_{SK}(c2c)			2.5	ns	
Part-to-Part Delay Skew	t_{SK}(p2p)			5.0	ns	

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5.7. Switching Characteristics - 2.5V Supply

(VDD1=2.5V± 10%, VDD2=2.5V± 10%, T_A=-40℃ to 125℃. Unless otherwise noted, Typical values are at **VDD1 = 2.5V, VDD2 = 2.5V,** T_A = 25℃)

Parameters	*Symbol*	*Min*	*Typ*	*Max*	*Unit*	*Comments*
Data Rate	DR	0		150	Mbps	
Minimum Pulse Width	PW			5.0	ns	
Propagation Delay	t_{PLH}	2.5	9.0	15	ns	See Figure 5.11 , C_L = 15pF
	t_{PHL}	2.5	9.3	15	ns	See Figure 5.11 , C_L = 15pF
Pulse Width Distortion $\lvert t_{PHL} - t_{PLH} \rvert$	PWD			5.0	ns	See Figure 5.11 , C_L = 15pF
Rising Time	t_r			5.0	ns	See Figure 5.11 , C_L = 15pF
Falling Time	t_f			5.0	ns	See Figure 5.11 , C_L = 15pF
Peak Eye Diagram Jitter	t_{JIT}(PK)		350		ps	
Channel-to-Channel Delay Skew	t_{SK}(c2c)			2.5	ns	
Part-to-Part Delay Skew	t_{SK}(p2p)			5.0	ns	

5.8. Typical Performance Characteristics

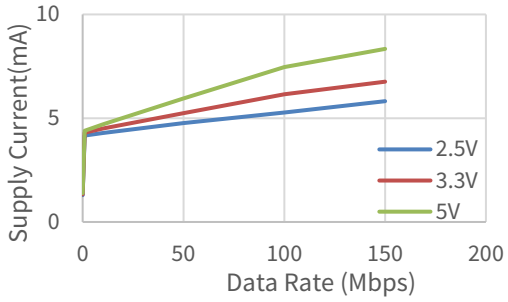


Figure 5.1 NSi8260 VDD1 Supply Current vs Data Rate

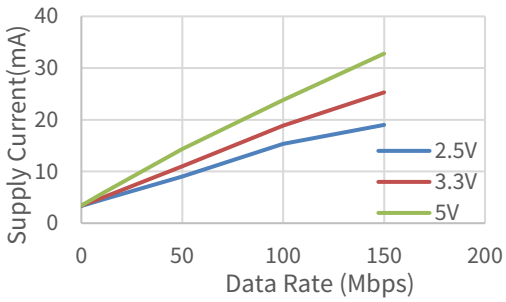


Figure 5.2 NSi8260 VDD2 Supply Current vs Data Rate

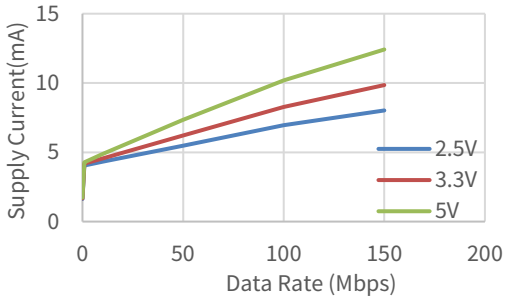


Figure 5.3 NSi8261 VDD1 Supply Current vs Data Rate

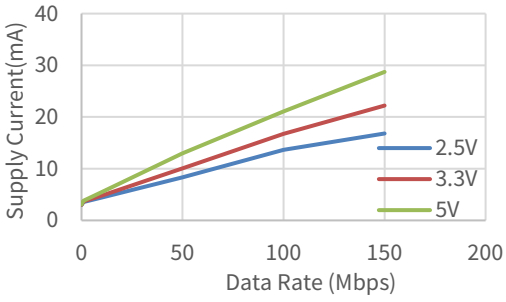


Figure 5.4 NSi8261 VDD2 Supply Current vs Data Rate

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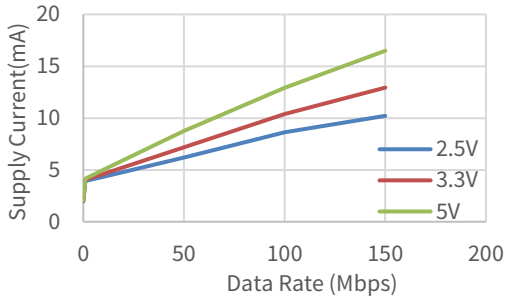


Figure 5.5 NSi8262 VDD1 Supply Current vs Data Rate

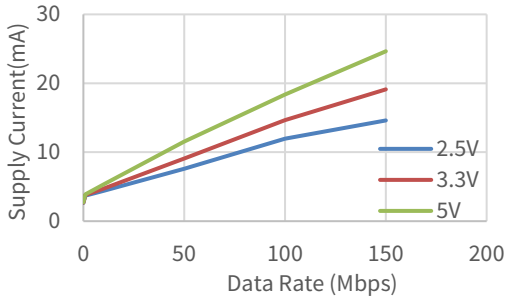


Figure 5.6 NSi8262 VDD2 Supply Current vs Data Rate

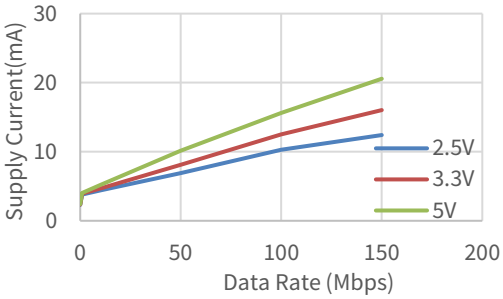


Figure 5.7 NSi8263 VDD1 Supply Current vs Data Rate

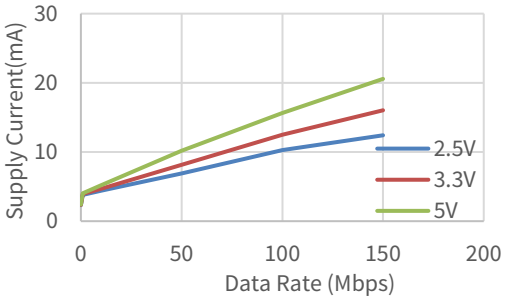


Figure 5.8 NSi8263 VDD2 Supply Current vs Data Rate

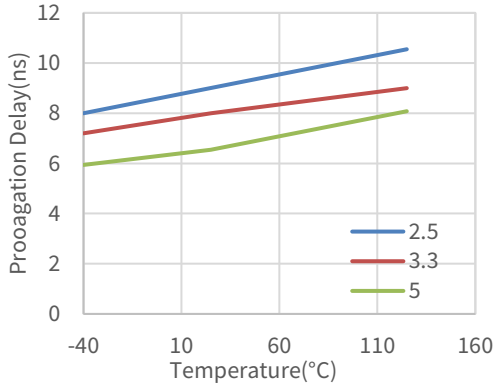


Figure 5.9 Rising Edge Propagation Delay Vs Temp

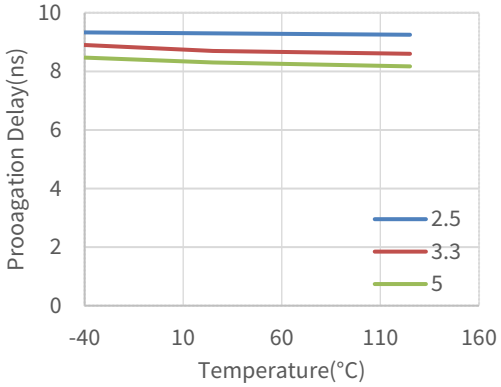


Figure 5.10 Falling Edge Propagation Delay Vs Temp

5.9. Parameter Measurement Information

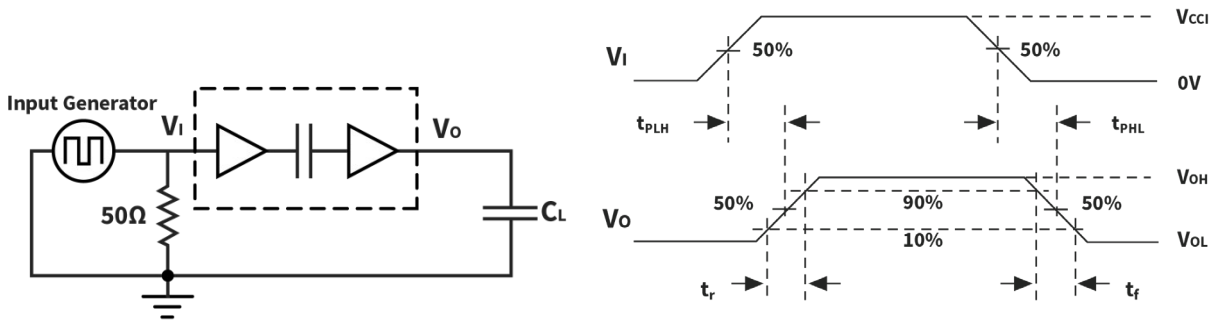


Figure 5.11 Switching Characteristics Test Circuit and Waveform

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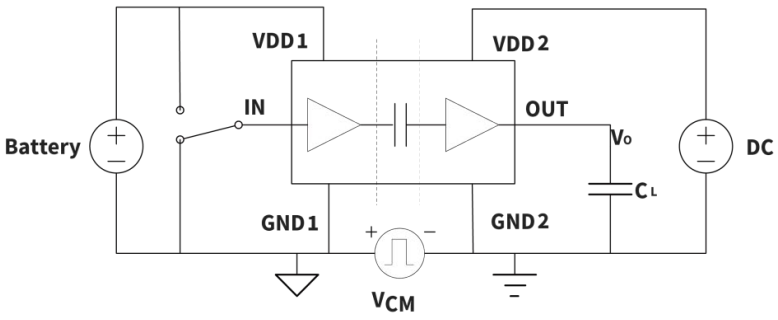


Figure 5.12 Common-Mode Transient Immunity Test Circuit

6. High Voltage Feature Description

6.1. Insulation and Safety Related Specifications

Description	*Test Condition*	*Symbol*	*Value*	*Unit*
			SOP16 (300mil)	
Min. External Air Gap (Clearance)		CLR	8	mm
Min. External Tracking (Creepage)		CPG	8	mm
Distance through the Insulation		DTI	28	um
Comparative Tracking Index	DIN EN 60112 (VDE 0303-11)	CTI	>600	V
Material Group	IEC 60112		I	
Installation Classification per DIN VDE 0110				
For Rated Mains Voltage ≤ 150Vrms			I to IV	
For Rated Mains Voltage ≤ 300Vrms			I to IV	
For Rated Mains Voltage ≤ 600Vrms			I to IV	
For Rated Mains Voltage ≤ 1000Vrms			I to III	
Insulation Specification per DIN EN IEC 60747-17 (VDE 0884-17) :2021-10[1)]				
Climatic Category			40/125/21	
Pollution Degree	per DIN VDE 0110, Table 1		2	
Maximum Working Isolation Voltage	AC voltage	V_{IOWM}	1500	V_{RMS}
	DC voltage		2121	V_{DC}
Maximum Repetitive Isolation Voltage		V_{IORM}	2121	V_{peak}

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Description	Test Condition	Symbol	Value	Unit
Input to Output Test Voltage, Method B1	$V_{ini.b} = V_{IOTM}$, $V_{pd(m)} = V_{IORM} \times 1.875$, $t_{ini} = t_m = 1$ sec, $q_{pd} \leq 5$ pC, 100% production test	$V_{pd(m)}$	3977	V_{peak}
Input to Output Test Voltage, Method A. After Environmental Tests Subgroup 1	$V_{ini.a} = V_{IOTM}$, $V_{pd(m)} = V_{IORM} \times 1.6$, $t_{ini} = 60$ sec, $t_m = 10$ sec, $q_{pd} \leq 5$ pC	$V_{pd(m)}$	3394	V_{peak}
Input to Output Test Voltage, Method A. After Input and Output Safety Test Subgroup 2 and Subgroup 3	$V_{ini.a} = V_{IOTM}$, $V_{pd(m)} = V_{IORM} \times 1.2$, $t_{ini} = 60$ sec, $t_m = 10$ sec, $q_{pd} \leq 5$ pC	$V_{pd(m)}$	2545	V_{peak}
Maximum Transient Isolation Voltage	t = 60 sec	V_{IOTM}	8000	V_{peak}
Maximum Surge Isolation Voltage	Test method per IEC62368-1, 1.2/50us waveform, $V_{TEST} = 1.6 \times V_{IOSM}$	V_{IOSM}	6250	V_{peak}
Isolation Resistance	V_{IO} = 500 V, $T_A = T_S$	R_{IO}	$>10^9$	Ω
	V_{IO} = 500 V, 100 °C $\leq T_A \leq$ 125 °C		$>10^{11}$	Ω
Isolation Capacitance	f = 1MHz	C_{IO}	1.2	pF
Insulation Specification per UL1577				
Withstand Isolation Voltage	$V_{TEST} = 1.2 \times V_{ISO}$, t = 1 sec, 100% production test	V_{ISO}	5000	V_{rms}

1) This coupler is suitable for “safe electrical insulation” only within the safety ratings. Compliance with the safety ratings shall be ensured by means of suitable protective circuits.

6.2. Safety-Limiting Values

Reinforced isolation safety-limiting values as outlined in VDE 0884-17 of NSi826xC-DSWR

Description	Test Condition	Value	Unit
Safety Supply Power	$R_{\theta JA}$ = 60.3 °C/W, T_J = 150 °C, T_A = 25 °C	2073	mW
Safety Supply Current	$R_{\theta JA}$ = 60.3 °C/ W, V_I = 5V, T_J = 150 °C, T_A = 25 °C	414	mA
Safety Temperature[2)]		150	°C

1) Calculate with the junction-to-air thermal resistance, $R_{\theta JA}$, of SOP16(300mil) package (***Thermal Information Table***) which is that of a device installed on a low effective thermal conductivity test board (1s) according to JESD51-3.

2) The maximum safety temperature has the same value as the maximum junction temperature (T_J) specified for the device.

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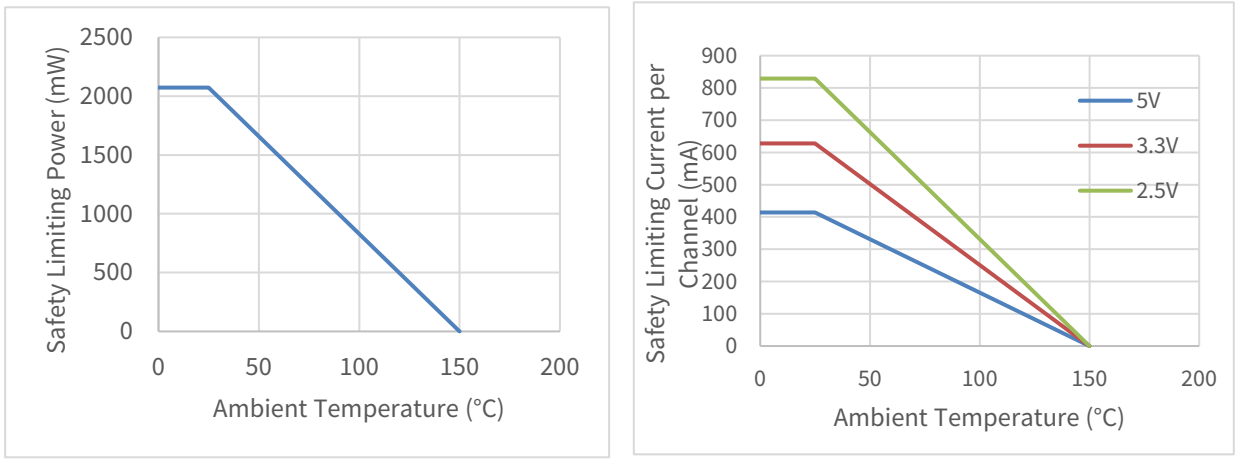


Figure 6.1 NSi826xC-DSWR Thermal Derating Curve, Dependence of Safety Limiting Values with Case Temperature per DIN EN IEC 60747-17 (VDE 0884-17)

6.3. Regulatory Information

The NSi826xC-DSWR are approved or pending approval by the organizations listed in table.

CUL		***VDE***	***CQC***
UL 1577 Component Recognition Program[1]	Approved under CSA Component Acceptance Notice 5A	DIN EN IEC 60747-17 (VDE 0884-17)	Certified according to GB4943.1-2022
Single Protection, 5000V_{rms} Isolation voltage	Single Protection, 5000V_{rms} Isolation voltage	Reinforce Insulation 2121Vpeak, V_{IOSM}=6250Vpeak	Reinforced insulation
File (E500602)	File (E500602)	File (40052820)	File (CQC20001264939)

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7. Function Description

7.1. Overview

The NSi826xC is a Six-channel digital isolator based on a capacitive isolation barrier technique. The digital signal is modulated with RF carrier generated by the internal oscillator at the Transmitter side. Then it is transferred through the capacitive isolation barrier and demodulated at the Receiver side.

The NSi826xC devices are high reliability six-channel digital isolator. The NSi826xC device is safety certified by UL1577 support 5kV$_{rms}$ insulation withstand voltages, while providing high electromagnetic immunity and low emissions at low power consumption. The data rate of the NSi826xC is up to 150Mbps, and the common-mode transient immunity (CMTI) is up to 250kV/us. The NSi826xC device provides digital channel direction configuration and the default output level configuration when the input power is lost. Wide supply voltage of the NSi826xC device support to connect with most digital interface directly, easy to do the level shift. High system level EMC performance enhance reliability and stability of use.

The NSi826xC has a default output status when VDDIN is unready and VDDOUT is ready as shown in Table 7.1, which helps for diagnosis when power is missing at the transmitter side. The output B follows the same status with the input A after powering up.

Table 7.1 Output status vs. power status

Input	***VDD1 status***	***VDD2 status***	***Output***	***Comment***
H	Ready	Ready	H	Normal operation.
L	Ready	Ready	L	
X	Unready	Ready	L(NSi826xC0) H(NSi826xC1)	The output follows the same status with the input after input side VDD is powered on.
X	Ready	Unready	X	The output follows the same status with the input after output side VDD2 is powered on.
Note: H=Logic high; L=Logic low; X=Logic low or logic high VDD1 is input side power; VDD2 is output side power.				

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7.2. OOK Modulation

NSi8266 is based on a capacitive isolation barrier technique and the digital signal is modulated with RF carrier generated by the internal oscillator at the transmitter side, as shown in Figure 7.1 to Figure 7.2, then it is transferred through the capacitive isolation barrier and demodulated at the receiver side. The modulation uses OOK modulation technique with key benefits of high noise immunity and low radiation EMI.

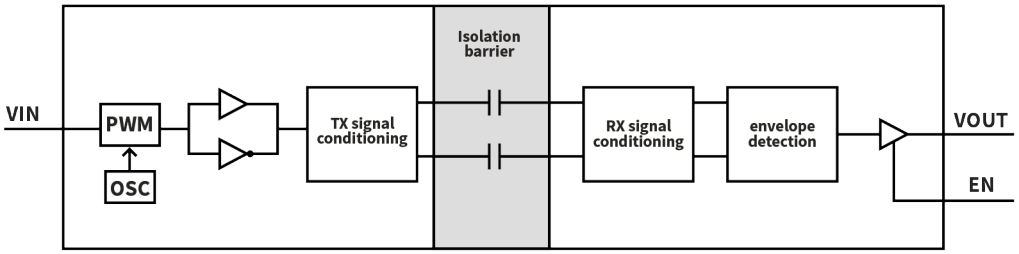


Figure 7.1 Single Channel Function Block Diagram

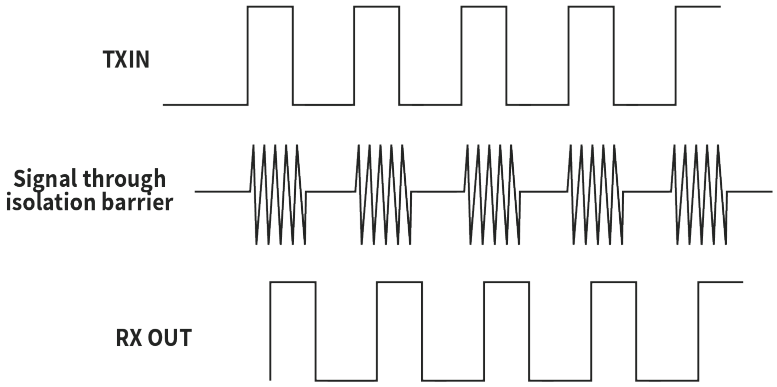


Figure 7.2 OOK Modulation

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8. Application Note

8.1. Typical Application Circuit

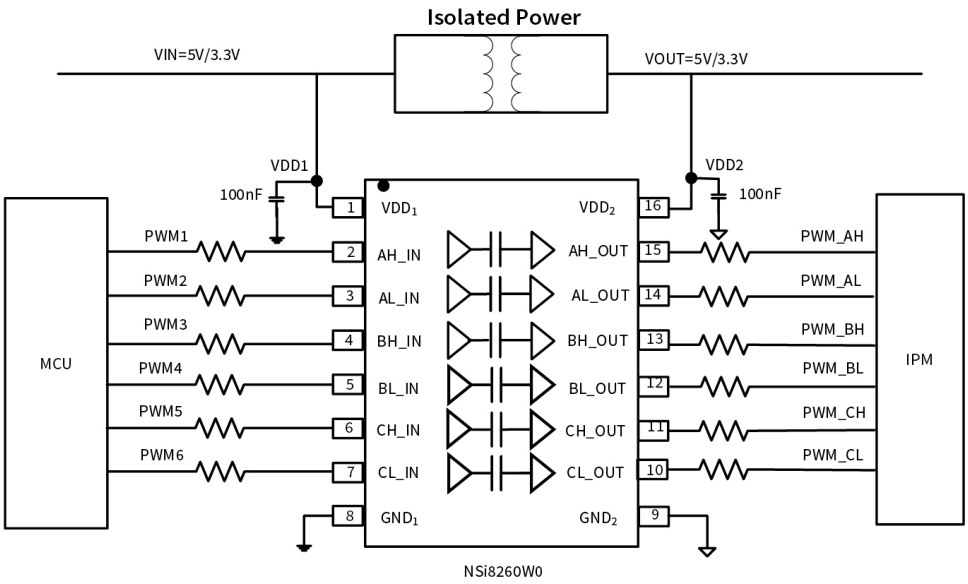


Figure 8.1 Typical PWM isolation circuit for IPM

8.2. PCB Layout

The NSi826xC requires a 0.1 µF bypass capacitor between VDD1 and GND1, VDD2 and GND2. The capacitor should be placed as close as possible to the package. Figure 8.2 to Figure 8.3 show the recommended PCB layout, make sure the space under the chip should keep free from planes, traces, pads and via. To enhance the robustness of a design, the user may also include resistors (50–300 Ω) in series with the inputs and outputs if the system is excessively noisy. The series resistors also improve the system reliability such as latch-up immunity.

The typical output impedance of an isolator driver channel is approximately 50 Ω, ±40%. When driving loads where transmission line effects will be a factor, output pins should be appropriately terminated with controlled impedance PCB traces.

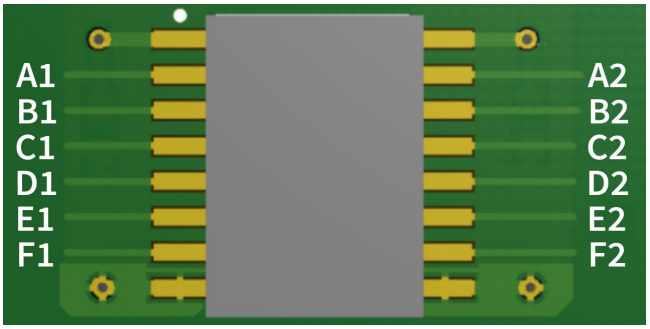


Figure 8.2 Recommended PCB Layout — Top Layer



Figure 8.3 Recommended PCB Layout — Bottom Layer

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8.3. High Speed Performance

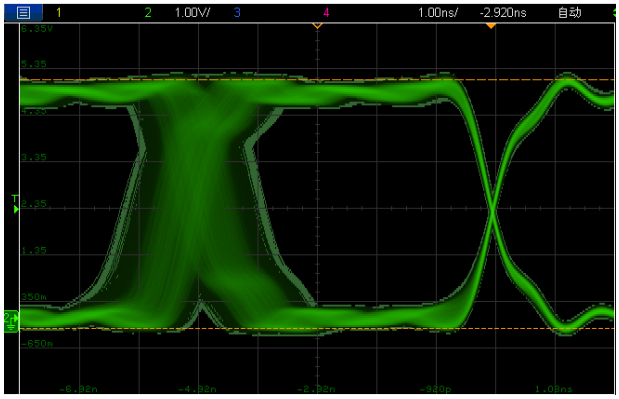


Figure8.4 Eye Diagram

8.4. Typical Supply Current Equations

The typical supply current of NSi826xC can be calculated using below equations. I_{DD1} and I_{DD2} are typical supply currents measured in mA, f is data rate measured in Mbps, C_L is the capacitive load measured in pF

NSi8260:

I_{DD1} = 0.19 *a1+1.45*b1+0.82*c1.

I_{DD2} = 1.36+ VDD2*f* C_L *c1*10^{-9}

When a1 is the channel number of default state input at side 1, b1 is the channel number of non-default state input at side 1, c1 is the channel number of switch signal input at side 1.

NSi8261:

I_{DD1} = 0.87 +1.26*b1+0.63*c1+ VDD1*f* C_L *c2*10^{-9}

I_{DD2} = 0.87 +1.26*b2+0.63*c2+ VDD2*f* C_L *c1*10^{-9}

When b1 is the channel number of non-default state input at side 1, c1 is the channel number of switch signal input at side 1, b2 is the channel number of non-default state input at side 2, c2 is the channel number of switch signal input at side 2.

NSi8262:

I_{DD1} = 0.87 +1.26*b1+0.63*c1+ VDD1*f* C_L *c2*10^{-9}

I_{DD2} = 0.87 +1.26*b2+0.63*c2+ VDD2*f* C_L *c1*10^{-9}

When b1 is the channel number of non-default state input at side 1, c1 is the channel number of switch signal input at side 1, b2 is the channel number of non-default state input at side 2, c2 is the channel number of switch signal input at side 2.

NSi8263:

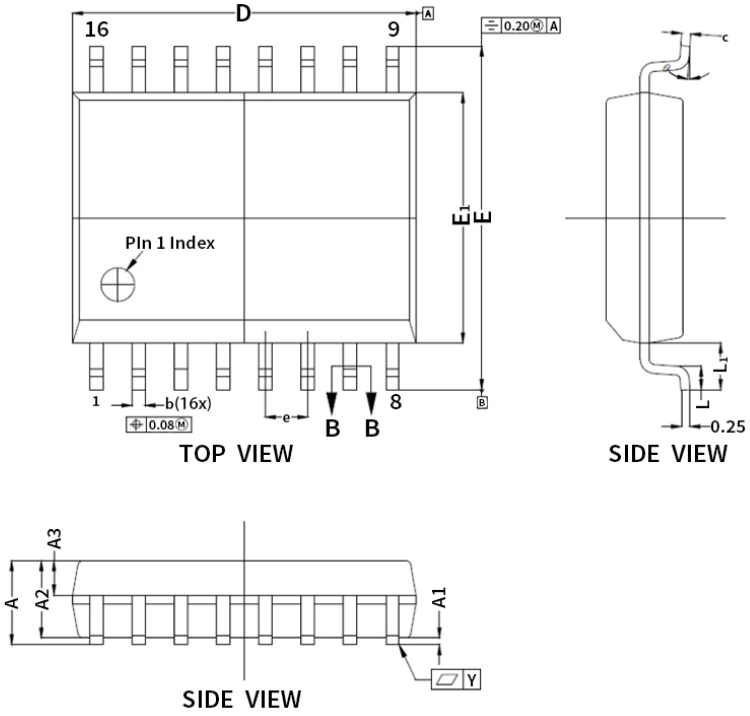
I_{DD1} = 0.87 +1.26*b1+0.63*c1+ VDD1*f* C_L *c2*10^{-9}

I_{DD2} = 0.87 +1.26*b2+0.63*c2+ VDD2*f* C_L *c1*10^{-9}

When b1 is the channel number of non-default state input at side 1, c1 is the channel number of switch signal input at side 1, b2 is the channel number of non-default state input at side 2, c2 is the channel number of switch signal input at side 2.

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9. Package Information



* CONTROLLING DIMENSION:MM

SYMBOL	MILLIMETER		
	MIN.	NOM.	MAX.
A	---	---	2.65
A1	0.10	---	0.30
A2	2.25	2.30	2.35
A3	0.97	1.02	1.07
b	0.35	---	0.43
c	0.23	---	0.32
D	10.20	10.30	10.40
E	10.10	10.30	10.50
E1	7.40	7.50	7.60
e	1.27 bsc		
L1	1.40 bsc		
L	0.55	---	0.85
Y	---	0.10	---
θ	0°	---	8°

Figure 9.1 SOP16(300mil) Package Shape and Dimension in millimeters

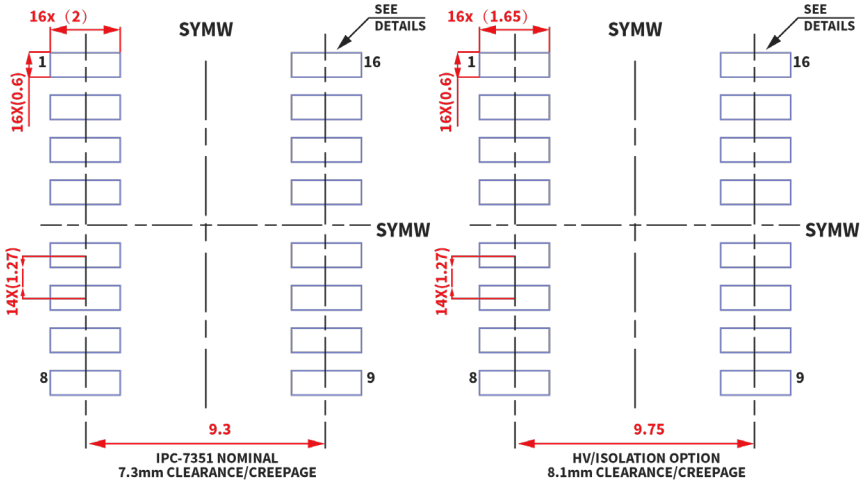


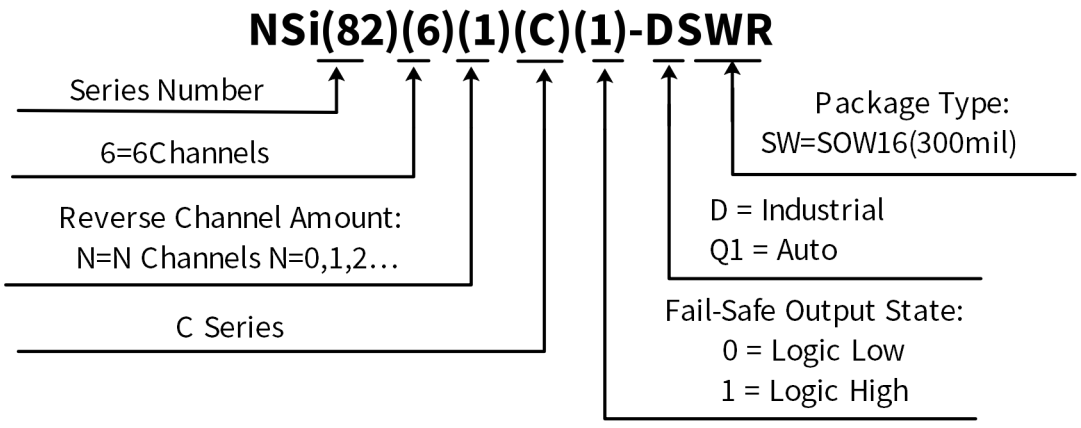
Figure 9.2 SOP16(300mil) Package Board Layout Example

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10. Order Information

Part Number	Isolation Rating (kV)	Number of side 1 inputs	Number of side 2 inputs	Max Data Rate (Mbps)	Default Output State	Temperature	MSL	Package Type	Package Drawing	SPQ
NSi8260C0-DSWR	5	6	0	150	Low	-40 to 125℃	3	SOP16 (300mil)	SOW16	1000
NSi8260C1-DSWR	5	6	0	150	High	-40 to 125℃	3	SOP16 (300mil)	SOW16	1000
NSi8261C0-DSWR	5	5	1	150	Low	-40 to 125℃	3	SOP16 (300mil)	SOW16	1000
NSi8261C1-DSWR	5	5	1	150	High	-40 to 125℃	3	SOP16 (300mil)	SOW16	1000
NSi8262C0-DSWR	5	4	2	150	Low	-40 to 125℃	3	SOP16 (300mil)	SOW16	1000
NSi8262C1-DSWR	5	4	2	150	High	-40 to 125℃	3	SOP16 (300mil)	SOW16	1000
NSi8263C0-DSWR	5	3	3	150	Low	-40 to 125℃	3	SOP16 (300mil)	SOW16	1000
NSi8263C1-DSWR	5	3	3	150	High	-40 to 125℃	3	SOP16 (300mil)	SOW16	1000
NOTE: All packages are RoHS-compliant with peak reflow temperatures of 260 °C according to the JEDEC industry standard classifications and peak solder temperatures.										

Part Number Rule:

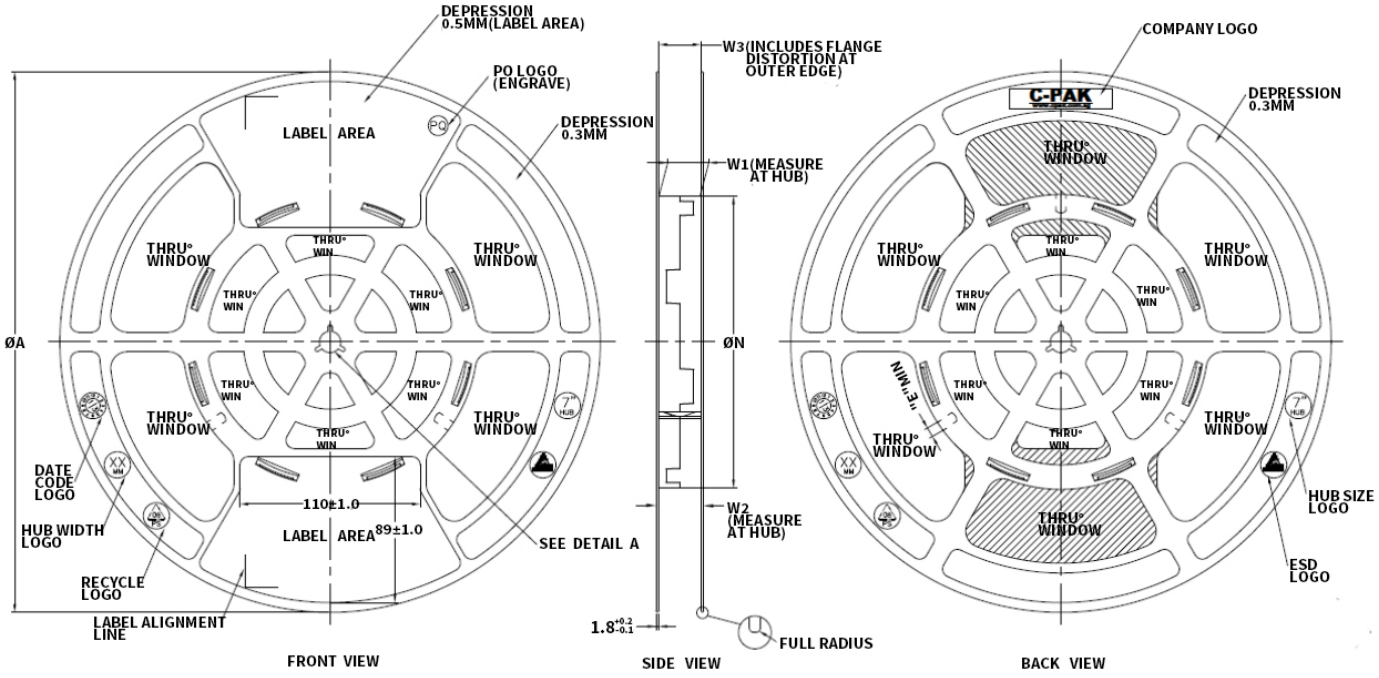


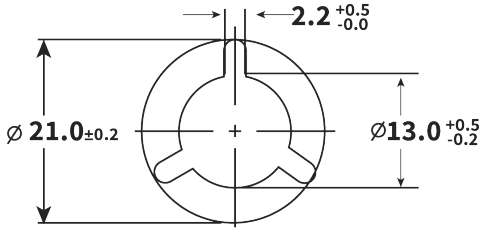
11. Documentation Support

Part Number	Product Folder	Datasheet	Technical Documents	Isolator selection guide
NSi826xC	tbd	tbd	tbd	tbd

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12. Tape and Reel Information





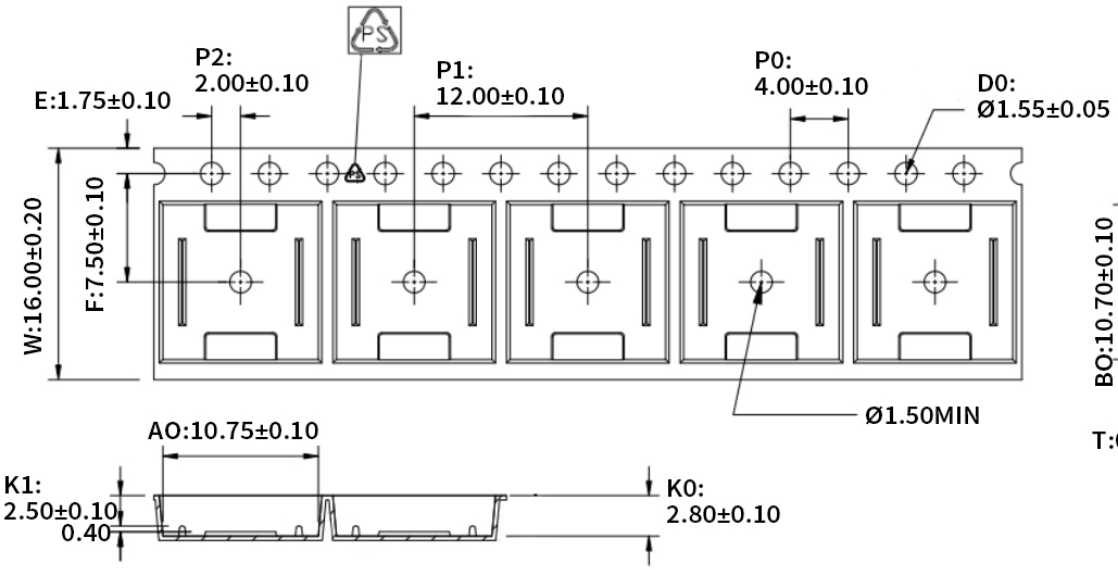
ARBOR HOLE
DETAIL A
SCALE: 3:1

PRODUCT SPECIFICATION						
TAPE WIDTH	ØA ±2.0	ØN ±2.0	W1	W2 (Max)	W3	E (MIN)
08MM	330	178	8.4 +1.5 -0.0	14.4	SHALL ACCOMMODATE TAPE WIDTH WITHOUT NTERFERENCE	5.5
12MM	330	178	12.4 +2.0 -0.0	18.4		5.5
16MM	330	178	16.4 +2.0 -0.0	22.4		5.5
24MM	330	178	24.4 +2.0 -0.0	30.4		5.5
32MM	330	178	32.4 +2.0 -0.0	38.4		5.5

SURFACE RESISTIVITY			
LEGEND	SR RANGE	TYPE	COLOUR
A	BELOW 10^{12}	ANTISTATIC	ALL TYPES
B	10^{6} TO 10^{11}	STATIC DISSIPATIVE	BLACK ONLY
C	10^{5} & BELOW 10^{5}	CONDUCTIVE(GENERIC)	BLACK ONLY
E	10^{9} TO 10^{11}	ANTISTATIC(COATED)	ALL TYPES

Figure 12.1 Reel Information (for all packages)

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1.10 sprocket hole pitch cumulative tolerance ±0.20.
2. Carrier camber is within 1 mm in 250 mm.
3. Material : Black Conductive Polystyrene Alloy.
4. All dimensions meet EIA-481 requirements.
5. Thickness: 0.30±0.05mm.
6. Packing length per 22" reel: 378 Meters.(Rewind N=122)
7. Component load per 13" reel: 1000 pcs.
8. Surface resistivity:10^5~ 10^{10}Ω/口

W	16.00±0.20
A0	10.75±0.10
B0	10.70±0.10
K0	2.80±0.10
K1	2.50±0.10

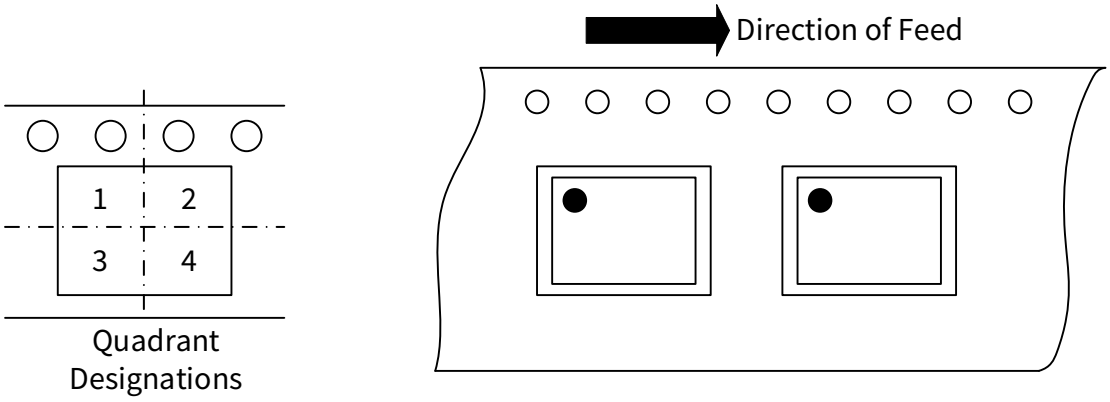


Figure 12.2 Tape Information of SOP16(300mil)

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13. Revision history

Revision	Description	Date
1.0	Initial version	2022/5/10
1.1	Delate “Isolation barrier life: >60 years”. Change DTI from 32 to 28, Change Min Storage Temperature to -65, add Junction Temperature, delate AEC-Q100 information.	2022/9/7
1.2	Update Regulatory Information and Safety Regulatory Approvals. Correct Typical Supply Current Equations, the icon in section 8 and MSL. Change the typical CMTI from 150 to 250 kV/us, change the minimum CMTI from 100 to 200 kV/us, change the maximum Data rate from 100 to 150Mbps.	2023/5/30

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